

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and indepth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

QUESTION 1: Why is star topology generally more expensive than bus topology, and how do they differ in performance and reliability?

ANSWER:

Introduction:

Star topology and bus topology are two common types of computer network arrangements. They differ in the way computers and other devices are connected. In a star topology, every device is connected separately to a central device, such as a switch or hub. In a bus topology, all devices share one main cable called the backbone cable.

Cost of Installation:

Star topology is generally more expensive than bus topology. It requires a central switch or hub, as well as a separate cable connection from every computer to the central device. As the number of devices increases, more cables and ports are needed. This increases the cost of equipment, installation, and maintenance.

Bus topology is cheaper because it uses one main backbone cable. Each computer is connected to this cable using a smaller connection. Since fewer cables and devices are required, bus topology is less expensive and easier to install, especially for small networks.

Reliability:

Star topology is more reliable because the failure of one computer or one connecting cable usually affects only that particular device. The remaining computers can continue communicating through the central switch or hub. However, if the central switch or hub fails, the entire network may stop working.

In bus topology, all devices depend on the same backbone cable. If the backbone cable is damaged or disconnected, communication across the whole network can be interrupted. Therefore, bus topology has a single point of failure that can affect every connected device.

Performance:

Star topology usually provides better performance than bus topology. The central switch manages the transmission of data and can send information directly to the intended device. This reduces collisions and allows several devices to communicate more efficiently.

In bus topology, all devices share the same communication medium. When many devices transmit data at the same time, collisions and network congestion may occur. As more devices are added, performance can decrease because all devices compete for access to the same backbone cable.

Conclusion:

Bus topology is less expensive and simpler to install, but it is less reliable and may experience reduced performance as the network grows. Star topology costs more because it requires a central device and separate cables, but it offers better performance, easier troubleshooting, and greater reliability. Therefore, star topology is usually preferred for modern networks, while bus topology may be suitable for small or temporary networks.

QUESTION 2: Why does mesh topology provide higher fault tolerance than star topology?

ANSWER:

Introduction:

Fault tolerance is the ability of a network to continue operating when one or more devices or connections fail. Mesh topology and star topology have different levels of fault tolerance because their devices are connected in different ways.

Mesh Topology and Multiple Paths:

In a mesh topology, each device is connected to several other devices. In a full mesh network, every device has a direct connection to every

other device. In a partial mesh network, devices have multiple connections, although not every device is connected directly to all the others.

These multiple connections create alternative paths for data. If one cable or connection fails, data can travel through another available route. As a result, communication can continue even when part of the network is damaged.

Reliability of Mesh Topology:

Mesh topology provides very high reliability and fault tolerance. For example, if one connection between two computers stops working, the computers may still communicate through other connected devices. This makes mesh topology suitable for important environments where network availability is essential, such as hospitals, banks, military systems, and data centers.

The main disadvantage is that mesh topology requires a large number of cables, network ports, and configuration settings. It is also more expensive and difficult to install and maintain, particularly when many devices are connected.

Fault Tolerance in Star Topology:

Star topology has moderate fault tolerance. If one computer or one cable connected to the central switch fails, only that individual device is affected. All other devices can normally continue working. This is an advantage over bus topology because one faulty computer does not usually bring down the entire network.

However, the central switch or hub is a critical point in the network. If it fails, all connected devices may lose communication. This means that star topology does not provide as many alternative paths as mesh topology.

Comparison:

Mesh topology is more fault tolerant because it has multiple routes for data transmission. Star topology depends mainly on one central

device, so its reliability depends on the condition of that switch or hub. A backup central device can improve the reliability of a star network, but it still generally does not provide the same level of redundancy as mesh topology.

Conclusion:

Mesh topology is more robust and reliable than star topology because its multiple connections allow data to be rerouted when a link fails. Star topology is easier and cheaper to manage, but failure of its central switch or hub can stop the entire network. Therefore, mesh topology is the better choice when maximum reliability and continuous communication are required.

QUESTION 3: Why is bus topology simpler and easier to install than mesh topology?

ANSWER:

Introduction:

The complexity of installing a network depends on the number of devices, cables, ports, and connections required. Bus topology and mesh topology differ greatly in this respect. Bus topology has a simple structure, while mesh topology has a large number of interconnections.

Structure of Bus Topology:

Bus topology uses one main backbone cable. All computers and network devices are connected to this shared cable. Because the network has a straightforward arrangement, fewer cables and connection points are needed.

The installation process is relatively simple. The backbone cable is placed across the required area, and each device is attached to it. Terminators are usually placed at both ends of the cable to prevent signals from reflecting back through the network.

Advantages of Bus Installation:

Bus topology is suitable for small networks because it requires less equipment and less cabling. It can be installed quickly and does not require a central switch or hub. The cost of materials and labor is also lower than in more complex topologies.

The simple design makes bus topology useful in situations where only a few devices need to be connected. It is also easier to understand for beginners because all devices share one common communication line.

Structure of Mesh Topology:

Mesh topology is much more complicated. Every device must have connections to several other devices. In a full mesh topology, each device is directly connected to every other device. For example, if a network has five devices, many separate links are required. If more devices are added, the number of connections increases rapidly.

This requires more cables, network ports, network interface equipment, and configuration work. Technicians must plan and test each connection carefully to ensure that the network works correctly.

Disadvantages of Mesh Installation:

Mesh topology takes more time and money to install. The large number of connections can make the network physically difficult to arrange. It can also be difficult to identify and repair a faulty connection because many cables may be connected to each device.

Although mesh topology provides excellent reliability, its installation is not practical for every network. It is usually selected when fault tolerance is more important than low cost and simplicity.

Conclusion:

Bus topology is simpler and easier to install because it uses a single backbone cable and requires fewer connections. Mesh topology is more complex because devices are connected through multiple links.

Therefore, bus topology is better for small, inexpensive, and simple networks, while mesh topology is more suitable for critical networks that need high reliability.

QUESTION 4: Why is hybrid topology more scalable than star topology?

ANSWER:

Introduction:

Scalability refers to the ability of a network to grow by adding more devices or sections without causing major problems. Hybrid topology and star topology can both be expanded, but hybrid topology generally provides greater flexibility.

Meaning of Hybrid Topology:

Hybrid topology is created by combining two or more different network topologies. For example, a network may combine star topology with bus topology, mesh topology, or another star network. The structure can be designed according to the needs of different departments, buildings, or locations.

This flexibility allows organizations to choose the most suitable topology for each part of the network. One department may use star topology for easy management, while another important section may use mesh connections for increased reliability.

Scalability of Hybrid Topology:

Hybrid topology can be expanded in sections. New switches, hubs, or network segments can be added without redesigning the entire network. If an organization opens a new office or adds a new department, that section can be connected to the existing network using the topology that best suits its requirements.

This makes hybrid topology suitable for large organizations, universities, hospitals, and companies with several branches. It

supports growth while allowing different parts of the network to operate according to their own needs.

Limitations of Star Topology:

Star topology is also scalable to a certain extent. New devices can be added by connecting them to the central switch or hub. This is easier than adding devices to a bus network. However, the central device has a limited number of ports and a limited processing capacity.

When all the ports are used, an additional switch or hub must be purchased and connected. If the network becomes very large, the central structure may become difficult to manage or may create a performance bottleneck. The network also depends heavily on the central devices.

Comparison:

Hybrid topology provides more flexibility because it does not depend on one fixed arrangement. It can combine different designs and can be expanded according to the physical layout and operational requirements of the organization. Star topology has a simpler structure, but its expansion is limited by the capacity of its central switch or hub.

Conclusion:

Hybrid topology is more scalable because it allows different topologies to be combined and new network sections to be added flexibly. Star topology is easier to install and manage for small and medium-sized networks, but it may require additional hardware when its central device reaches its limits. Therefore, hybrid topology is usually the better choice for large and growing networks.

QUESTION 5: Which network topology is easiest to maintain, and why?

ANSWER:

Introduction:

Network maintenance includes monitoring the network, finding faults, replacing damaged equipment, managing cables, and ensuring that devices can communicate properly. The ease of maintenance depends on the network's design and the number of connections it contains.

Maintenance of Star Topology:

Star topology is generally the easiest topology to maintain. Each device has a separate connection to the central switch or hub. If one computer stops working, the technician can inspect that device and its individual cable without disturbing the other computers.

Faults can usually be identified and isolated quickly. Adding or removing a device is also relatively easy because the technician only needs to work with one connection. The central switch can often provide useful information about connected devices and network activity.

Maintenance of Mesh Topology:

Mesh topology is highly reliable but difficult to maintain. It contains many cables and connections, so troubleshooting can take considerable time. A technician may need to check several possible paths to determine where a problem has occurred.

The large number of links also makes cable management difficult. Although one failed connection may not stop the network, it still needs to be repaired to preserve the network's redundancy and reliability. Mesh topology therefore requires skilled administration and careful documentation.

Maintenance of Bus Topology:

Bus topology is harder to maintain than star topology because all devices share one backbone cable. A problem in the backbone can affect the entire network. Finding the exact location of a break or loose connection may be difficult because communication problems can occur across the whole network.

The network may also become slower when many devices are active. Adding or removing devices can cause further problems if the cable is not terminated or connected correctly.

Maintenance of Hybrid Topology:

Hybrid topology has moderate maintenance complexity. It may be easy to maintain each individual section, but the complete network can be difficult to manage because different topologies may be connected together.

A fault in one section may require technicians to understand both that section and the way it connects to the rest of the network. Proper monitoring, documentation, and skilled administration are important for keeping a hybrid network functioning efficiently.

Conclusion:

Star topology is usually the easiest to maintain because each device has an independent connection and faults can be isolated quickly. Mesh topology is the most complex to troubleshoot because it has many connections. Bus topology is difficult to maintain because a single backbone failure can affect the entire network, while hybrid topology requires careful management because it combines different network structures. Therefore, star topology is generally the best choice when easy maintenance is a priority.

OVERALL CONCLUSION:

Each network topology has its own advantages and disadvantages. Bus topology is inexpensive and easy to install, but it is less reliable and may become slow as more devices are added. Star topology

provides good performance, reliability, and easy maintenance, although it costs more and depends on a central switch or hub. Mesh topology offers the highest fault tolerance because it provides multiple paths for data, but it is expensive and difficult to install. Hybrid topology is flexible and highly scalable because it combines two or more topologies. The best topology depends on the size, budget, reliability requirements, and future growth of the network.